MIL-R-36540B
5 November 1985
SUPERSEDING
MIL-R-36540A
20 November 1975

MILITARY SPECIFICATION

RAZOR, SURGICAL PREPARATION (FOR FIELD USE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a straight type surgical preparation razor assembly, suitable for emergency field use.

1.2 Classification. The razor assemblies covered by this specification shall be of the following styles, as specified (see 6.1):

Style 1 - Razor with metal rib.
Style 2 - Razor with plastic rib.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following sppecifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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MILITARY

MIL-P-116 - Methods of Preservation.

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-151 - Metal, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

DRAWING

Defense Personnel Support Center (DPSC)

23006 - Razor, Surgical Preparation

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicittion.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 - Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E 92 - Tests for Vickers Hardness of Metallic Materials.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Blade. The blade shall be fabricated of a high chromium stainless steel with a carbon content of no less than 0.50 percent. The beveled edge shall be uncoated or coated with polytetrafluoroethylene at the option of the contractor.

3.1.2 Rib. The rib of style 1 blade shall be fabricated from corrosion-resisting steel, or low carbon steel, nickel plated. The rib of the style 2 blade shall be made of a rigid plastic, with or without color. The plastic rib shall withstand fracture when tested as specified in 4.4.3.

3.1.3 Handle. The handle of the blade shall be made of rigid plastic with or without color. The handle shall withstand fracture when tested as specified in 4.4.3. The rolled flanged eyelet in the handle shall be made of stainless steel or shall be made of brass, plated or unplated.

3.2 Style, design and dimensions. The razor assembly shall be a straight surgical preparation razor suitable for emergency field use. The razor shall consist essentially of a blade with rib, and a handle. The blade shall have a single sharp cutting edge with a securely attached rib on the opposite edge. The handle shall be securely attached to blade by means of an eyelet. The eyelet shall hold the parts joined with sufficient tension (and without movement of eyelet itself) for the blade to fold into the handle to protect the blade, and to stay in an open position when in use. The design and dimensions of the razor shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing No. 23006 for style 1 or 2, as specified (see 6.1). The requirements specified herein for one style shall apply to that style only, and shall not be interchangeable with the other style. Style 1 razor shall have a metal rib and a plastic handle. Style 2 razor shall have a plastic rib, and a plastic handle with a locking detent.

3.2.1 Drawing. The drawing forming a part of this specification is a specification control drawing. The supplier shall be responsible for preparing his own shop drawing. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

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3.3 Finish.

3.3.1 Blade. The cutting edge of the blade shall be bilaterally ground and honed to razor sharpness. The final honed bevels shall be flat and smooth and shall have an included bevel angle of 14 to 24 degrees. The cutting edge except for the extreme 1/8 inch at the back end, shall be free from micro-cracks, and not more than 5 percent of the edge length shall show nicks, notches, or unhoned portions when examined as specified in 4.3.1. The remainder of the blade shall have a smooth, uniform finish, free from coarse grinding marks and pits. The entire blade shall be free from corrosion and scale.

3.3.2 Rib. Style 1 razor shall have a smooth bright finish rib. Style 2 razor shall have a smooth or matte finish rib.

3.3.3 Handle. The handle shall have a smooth or matte finish. The eyelet shall be smoothly peened.

3.4 Hardness. The blade shall have a Rockwell Superfical Hardness number of 15N 85.0 to 92.5 (Vickers Hardness Number 500 to 860) when tested as specified in 4.4.2.

3.5 Workmanship. Razor shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, plastic, and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number or contractor's purchase order number, the name of the manufacturer or supplier, the NSN, the Item Identification (optional), the lot size, the sample size, the name of the material, the lot number, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables I, II, and III (see 4.2.1, 4.2.2, 4.3) and 4.3.1.

4.2.1 Sampling for examination. A random sample of razors shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptable Quality Level (AQL) shall be as indicated in Table I. Unit of product for examinaion purposes shall be one razor.

TABLE I. Examination Criteria.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.5
Major B	S-2	2.5
Major C	S-4	2.5
Minor A	II	4.0
Minor B	S-2	6.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one razor except as otherwise indicated in the table.

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TABLE II. Tests.

Component	Characteristic	Rqmt.	Test Procedure	Inspection level
Steel	Chemical composition	3.1	4.4.1	See 4.2.3
Blade	Hardness	3.4	4.4.2	S-2
Razor	Fracture	3.1.2 and 3.1.3	4.4.3	S-1

4.2.3 Sampling of raw materials. A sufficient quantity of each lot of steel used in the fabrication of the components of the razor shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The razor assembly shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in Table III.

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TABLE III. Classification of defects.

Categories	Defects
Major A	
101	Style and design not as specified.
102	Cutting blade not firmly and securely held in rib.
103	Rib not properly secured to handle.
104	Cutting edge of blade not bilaterally ground and honed to razor sharpness.
105	Bevels of cutting edge not flat and uniformly sharpened.
106	Blade not free of corrosion and scale.
107	Razor does not open and close properly.
Major B	
151	Cutting edge has more than 5 percent of edge length with nicks, notches or unhoned portions (see 4.3.1).
152	Bevel angle of cutting edge not 14 to 24 degrees.
Major C	
171	Cutting edge (except for the extreme 1/8 inch back end) not free from micro-cracks (see 4.3.1).
Minor A	
201	Finish on blade not smooth and uniform.
202	Rib for style 1 not smooth bright finish.
203	Rib for style 2 not smooth or matte finish.
204	Eyelet not smoothly peened and free from rough edges.
Minor B	
251	Dimensions (other than 152) not as specified.

4.3.1 Microscopic examination of razor blade cutting edge. Prior to examination, any oil, grease, and lint shall be removed from the cutting edge of the blade by any suitable means which will not damage the blade edge. For coated blades, this step shall be performed prior to coating the blade. The blade shall be examined for possible imperfections using an 80-100 power microscope with adequate illumination.

4.4 Test procedures.

4.4.1 Chemical analysis. In the absence of certificates of quality, chemical analysis of the individual components of the razor assembly shall be determined in accordance with Method 111 or 112 of FED-STD-151. In the event of conflict, Method 111 shall govern.

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4.4.2 Hardness. Hardness shall be determined in accordance with American Society for Testing and Materials (ASTM) Designation E 18 (using the 15N scale, a 15 kilogram-force (kgf) load, and superficial diamond penetrator) or in accordance with ASTM Designation E 92 (using a 5 kgf load). Impressions shall be taken as close to the cutting edge as practicable, but not on the beveled portion. The average of three impressions shall be considered the hardnesss value of the blade.

4.4.3 Fracture test. The closed razor shall be placed flat on a table 36 inches above a concrete floor and shall be dropped freely from the table to the floor. The razor shall then be examined for fracture of the plastic components. The presence of scuff marks or a slightly marred finish as a result of the test shall not be cause for rejection.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Each razor shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Each razor shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit package. Each razor shall be packaged in a clear snug fitting polyethylene container having a nominal wall thickness of 0.003 inches. Excess air shall be evacuated from the container and the closure effected by heat sealing. The container shall be waterproof and airtight.

5.1.1.5 Intermediate package. Fifty razors shall be packaged in a box conforming to PPP-B-566, PPP-B-676, or PPP-B-636, type CF, class domestic. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for razors shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Sixteen intermediate packages (800 razors) shall be packed in a box designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to tne applicable paragraphs of the apppendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Sixteen intermediate pakages (800 razors) shall be packed in a box designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, type CF, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for razors shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Razors, packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Razors, packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit package, intermediate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

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6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-926-2089	RAZOR, SURGICAL PREPARATION, Straight Type, Small

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6515-3248

Location: S2-9-6

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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